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weitsicht: Python framework and utilities for monoplotting and projecting coordinates using geo-referenced imagery.

Martin Wieser [1¶]

1 Department of Geodesy and Geoinformation, TU Wien, Austria ¶ Corresponding author

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Software

- Review
- Repository
- Archive

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Summary

weitsicht is an Apache-2.0 Python library for working with geo-referenced imagery by connecting image pixel coordinates to 3D world coordinates. It implements (i) projection of 3D coordinates into a camera model and (ii) single-image ray/surface intersection ("monoplotting") to map image measurements to 3D points using terrain representations such as planes, digital surface models (rasters), or 3D meshes. **weitsicht** is designed to leverage direct georeferencing (e.g., IMU/GNSS/INS metadata from drones) or camera poses computed externally (e.g., from photogrammetry/SfM pipelines) and to provide a compact, GIS-oriented API with coordinate reference system (CRS) handling via `pyproj` (PROJ contributors, 2025), raster handling via `rasterio` (Gillies & others, 2013) (and therefore GDAL (GDAL/OGR contributors, 2025)), and numerical foundations in `NumPy` (Harris et al., 2020). The source code for **weitsicht** has been archived to zenodo with the associated DOI: https://doi.org/10.5281/zenodo.23566514

Statement of need

Aerial imagery is widely used in wildlife monitoring and ecological surveys, where researchers often require rapid, repeatable conversion of image detections into mapped coordinates for abundance estimation and spatial modelling. **weitsicht** draws on several research and software roots; one major origin is the geometric foundation developed for **WISDAMapp (Wildlife Imagery Survey — Detection and Mapping)**(WISDAM team, 2024), a GUI used by marine and terrestrial megafauna researchers to digitize objects in images, enrich them with metadata, and map them to real-world coordinates without requiring deep photogrammetry expertise. In these operational settings, an approximate direct georeference (from INS/IMU+GNSS logs or image EXIF/XMP) is frequently sufficient, while full 3D reconstruction with Structure-from-Motion (SfM) can be unnecessarily complex, time-consuming, or prohibitively expensive for non-specialists. During a major refactoring leading up to the first public release of WISDAMapp, its geometric core functions were separated from the GUI and, together with functionality from other projects, formed **weitsicht** .

Beyond ecology, single-image mapping is also valuable in airborne archaeology and rapid documentation, where an INS solution may provide direct georeferencing for each exposure but only sparse or single images are available. **weitsicht** supports these workflows by extracting pose and camera information from metadata, applying camera distortion models, and intersecting rays with user-provided terrain models.

For offshore surveys (a common operating environment for WISDAMapp users), SfM can fail due to low texture, specular reflections, and moving water surfaces; in contrast, direct georeferencing combined with monoplotting can still provide usable footprints, center points, and mapped measurements.

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weitsicht also supports mapping pixels onto existing 3D surfaces, motivated in part by the **INDIGO** graffiti documentation project (INDIGO project, 2022), where image-based annotations can be projected onto a 3D model for spatial analysis and dissemination.

State of the field

A broad ecosystem of photogrammetry and SfM tools exists, including open-source packages such as COLMAP (Schönberger & Frahm, 2016), OpenMVG (Moulon et al., 2016), MicMac (Pierrot-Deseilligny & Cléry, 2011), OpenDroneMap (OpenDroneMap contributors, 2026), and commercial suites such as Agisoft Metashape and Pix4D. These systems excel at estimating camera geometry from multi-view imagery and producing dense reconstructions, but they typically present a steep learning curve for students and domain scientists and focus on the reconstruction stage rather than lightweight downstream operations on already geo-referenced images (e.g., computing footprints, intersecting rays with arbitrary terrain models, or projecting mapped features back into images for re-sighting workflows).

At a lower level, computer-vision libraries such as OpenCV (Bradski, 2000) provide camera models, distortion, and projection primitives. However, assembling a complete monoplotting pipeline from such building blocks (including robust CRS transformations, metadata-driven direct georeferencing, and interchangeable terrain backends) remains substantial work for non-expert users. **weitsicht** fills this gap by packaging established photogrammetric geometry into a small set of composable abstractions aimed at geospatial workflows.

Software design

weitsicht is organized as modular subpackages:

- `weitsicht.camera`: perspective camera models (including an OpenCV-style distortion model) used for projection and ray generation.
- `weitsicht.image`: image models combining camera intrinsics, exterior orientation, and CRS context, exposing `project_*` and `map_*` methods for 3D-2D transformations.
- `weitsicht.mapping`: interchangeable mappers that define the terrain/surface used for ray intersections, including a horizontal plane mapper, a raster/DSM mapper based on `rasterio` (Gillies & others, 2013) (and therefore GDAL (GDAL/OGR contributors, 2025)), and a mesh mapper based on `trimesh` (Dawson-Haggerty & trimesh contributors, 2026).
- `weitsicht.metadata`: Utilities to estimate camera intrinsics and extract exterior orientation and coordinate reference system (CRS) information from EXIF/XMP metadata (e.g., from drone payloads recording IMU/GNSS/INS data). The current implementation supports metadata tags produced by ExifTool but can be easily extended to other tag naming conventions, such as those used by Exiv2 backends, through a common base tag class.
- `weitsicht.transform`: CRS and rotation utilities with a consistent "always x/y" convention to reduce common user errors when transforming between geographic and projected coordinate systems.

Modularity is an explicit design goal: new camera models (e.g., fisheye or panoramic/360° cameras) and new mapping backends (e.g., point-cloud ray intersections or tiled/streamed raster sources such as cloud-optimized GeoTIFFs) can be implemented by extending the corresponding camera or mapper interfaces while keeping the image-level API stable.

This design enables users to start from minimal inputs (an image, metadata, and a terrain model) and progress to mapped 3D points, footprints, or back-projected image coordinates with clear error handling and typed results. Advanced users can extend the library by implementing

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additional camera models, metadata backends, or mapping surfaces while reusing the same projection and CRS infrastructure.

Python was chosen due to its widespread adoption in geospatial science and its extensive ecosystem of scientific libraries. Although computational performance may be lower than that of compiled languages, only basic performance optimizations have been implemented so far, leaving room for future improvements.

Research impact statement

By separating the geometric foundation from the WISDAM GUI, **weitsicht** provides reusable core functionality for multiple applied research domains. Within WISDAMapp (WISDAM team, 2024), it supports workflows for wildlife monitoring (e.g., marine megafauna and dugong surveys) using drones and crewed aircraft, enabling researchers to map observations from geo-referenced imagery and to project mapped locations back into images to facilitate re-sighting and quality control. WISDAM was developed iteratively: earlier releases under the names `OceanMapper` (a GUI-less monoplotting prototype, see (Cleguer et al., 2021)) and `DugongDetector` (an initial, dugong-focused GUI) represent development stages of the same software that is now released as WISDAM. The workflow builds on UAV/aerial imagery survey methodology and detection-probability estimation (Hodgson et al., 2017). The software framework has been used in peer-reviewed research via using WISDAM, for example in a recent dugong habitat assessment using paired drone and in-water surveys (Said et al., 2025). (Digdo et al., 2025) report a drone survey for dugongs in a limited-resource context that combined manual analysis with AI-assisted detection using WISDAM. And (Raudino et al., 2026) used WISDAM for their review workflow in a wuthuga (dugong) abundance study using ariel imagery. The WISDAM app has been developed and applied in collaborations involving institutions including Edith Cowan University (Centre for Marine Ecosystems Research), Murdoch University (Centre for Sustainable Aquatic Ecosystems, Harry Butler Institute), TU Wien (photogrammetry and remote sensing), and James Cook University (TropWATER), and in projects with partners such as the Seychelles Islands Foundation, Insitu Pacific Inc., the Australian Antarctic Division and Queensland University of Technology as documented on the project website (WISDAM team, 2024).

In archaeological and cultural-heritage contexts, **weitsicht** supports direct-georeferencing and single-image mapping workflows relevant to projects such as it was implemented by ARAP (University of Vienna) (ARAP project, 2017) and INDIGO (INDIGO project, 2022), where efficient projection between images and 3D surfaces enables rapid documentation, spatial analysis, and dissemination without requiring a full SfM pipeline (Doneus et al., 2016; Verhoeven et al., 2013; Wieser et al., 2014, 2024). Parts of functionality developed in that projects are some of the roots beside `WISDAMapp` and where consequently enhanced and improved.

AI usage disclosure

Generative AI was used for assistance and then edited and verified by the author.

Tool used: ChatGPT (model GPT-5.2) for documentation, code, and CI/CD maintenance.

Scope of assistance: docstring harmonization; spelling and grammar checks in documentation; drafting the latest function for advanced GSD (Ground Sampling Distance) estimation; assistance creating CI/CD workflows for GitHub Actions.

I confirm as the author that human authors reviewed, edited and validated all AI-assisted outputs and made the core design decisions.

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